

Homework 10

40 points

1. Box A has 2 red, 3 blue. Box B has 4 red, 1 blue. You choose a box at random (equal chance) and draw one ball. It is red. What is the probability you picked Box B? **3 points**

2. A single die is rolled. What is the probability of getting an even number, given that a number greater than 3 is rolled? **3 points**

3. A DNA test has the following accuracy:

- True match rate 99%
- False match rate 1%

Only **1 in 500 people** could be the true match.

Someone matches. What is probability they are the true match? **4 points**

4. You have two coins: A fair coin and a double-headed coin. You pick a coin randomly and get Heads. Find the probability that the coin is double-headed.

3 points

5. An insurance company knows that a customer has a 2% chance of filing a \$10,000 claim this year. What is the expected payout per customer?

If the company wants an expected profit of \$200 per customer, what should be the premium? **3 + 2 = 5 points**

6. A slot machine costs \$2 per play. Payouts:

Prize	Amount	Probability
Jackpot	\$500	1/2000
Small win	\$20	1/50
No win	\$0	remainder

Find the expected net gain per play. **5 points**

7. A factory wants the sample average weight of packages to be within ± 0.2 kg of the true mean with probability 0.95. Package weights have standard deviation $\sigma = 1.5$ kg. Using CLT, how large should the sample n be? **6**

points

8. Let $X_1, \dots, X_{64}$ be independent and identically distributed with mean $\mu = 10$ and variance $\sigma^2 = 16$. Approximate the probability that the sample mean $\bar{X}$ exceeds 11. **5 points**

9. Ten lightbulbs lasted (in hours):

2800, 2900, 3050, 3150, 2700, 3000, 2990, 2850, 3100, 2950.

Find a **95% CI** for the mean lifetime (assume normality). **6 points**